Society for Computer Technology & Research
Pune Institute of Computer Technology, Pune

Department of Information Technology

LAB RECORD

Banking, Financial Services and Insurance (BFSI)

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Class:	SY-09
Semester:	IV
Course Code:	0405322
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Experiment 1: Study of BFSI Domain & Statistical Analysis

Aim

To study the BFSI domain and apply basic statistical techniques on banking data to understand data-driven financial decision-making.

Objectives

- Understand the structure and key components of the BFSI domain.
- Load and inspect a real-world banking dataset.
- Apply descriptive statistics — mean, median, standard deviation, min, max.
- Perform Pearson correlation analysis between key financial variables.
- Engineer a credit score category feature for grouped analysis.
- Derive meaningful financial insights from the statistical output.

Theory

The **BFSI (Banking, Financial Services and Insurance)** sector is one of the most data-intensive industries in the world. It encompasses:

- **Banking** – retail banking, corporate banking, loans, deposits, and payments.
- **Financial Services** – investment management, stock broking, mutual funds, and wealth management.
- **Insurance** – life insurance, general insurance, and health insurance.

Data science in BFSI relies heavily on statistical analysis for credit underwriting, fraud detection, risk assessment, and portfolio management. The core statistical tools used are:

- **Mean** – arithmetic average; gives the central value of a financial variable (e.g., average loan amount).
- **Median** – middle value; robust to outliers, preferred for skewed distributions like income.
- **Standard Deviation (σ)** – measures spread or variability; high σ in loan amounts indicates a diverse borrower profile.

- **Pearson Correlation (r)** – measures the linear relationship between two variables, ranging from -1 to $+1$.
  - $r > 0$: positive relationship (both increase together)
  - $r < 0$: negative relationship (one increases as other decreases)
  - $r \approx 0$: little to no linear relationship
- **Risk-based Pricing** – banks set interest rates based on borrower credit risk; customers with lower credit scores are charged higher rates to compensate for default risk.

Dataset Used

Dataset Name: `banking_dataset.xlsx`

Source: Synthetic banking dataset.

Size: 1000 rows × 12 columns

Key Columns:

- `Customer_ID` – Unique customer identifier
- `Age` – Customer age in years
- `City` – Customer's city (Mumbai, Delhi, Bangalore, Chennai, Hyderabad, Pune)
- `Annual_Income` – Annual income in INR
- `Credit_Score` – Credit score (range: 300–900)
- `Loan_Type` – Type of loan (Home, Personal, Car, Education)
- `Loan_Amount` – Loan amount sanctioned in INR
- `Interest_Rate` – Rate of interest (%)
- `Loan_Tenure_Years` – Repayment period in years
- `Transaction_Amount`, `Transaction_Type`, `Transaction_Date` – Recent transaction details

Procedure

1. Load the dataset `banking_dataset.xlsx` using the `pandas` library.
2. Inspect the dataset: print shape, column names, data types, and first 5 rows.

3. Compute descriptive statistics using `df.describe()` and individual aggregation functions.
4. Extract key statistics: mean, median, and standard deviation for Annual Income, Loan Amount, and Credit Score.
5. Select numerical columns and compute the Pearson correlation matrix using `df.corr()`.
6. Highlight three key correlation pairs: Income vs. Loan Amount, Credit Score vs. Interest Rate, and Credit Score vs. Loan Amount.
7. Engineer the `Credit_Category` feature by binning credit scores into four bands.
8. Plot a histogram of Annual Income with mean and median reference lines.
9. Plot a correlation heatmap of all numerical features.
10. Print a final summary of observations and results.

Program Code

```
# ============================================================
#  BFSI Lab Assignment 1
#  Study of BFSI Domain & Statistical Analysis on Banking Data
#  Name: Piyush Rajesh Ahir | Roll: 23104 | SY-09 | AY: 2025-26
# ============================================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

from google.colab import files
uploaded = files.upload()

df = pd.read_excel('banking_dataset.xlsx')

# STEP 1: Dataset Overview
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")
print(df.head())
print(df.dtypes)
```

```
# STEP 2: Descriptive Statistics
print(df.describe().round(2).to_string())
print(f"Mean   Annual Income : Rs {df['Annual_Income'].mean():,.2f}")
print(f"Median Annual Income : Rs {df['Annual_Income'].median():,.2f}")
print(f"Std    Annual Income : Rs {df['Annual_Income'].std():,.2f}")
print(f"Mean   Loan Amount   : Rs {df['Loan_Amount'].mean():,.2f}")
print(f"Mean   Credit Score  : {df['Credit_Score'].mean():.2f}")
print(f"Mean   Interest Rate : {df['Interest_Rate'].mean():.2f} %")

# STEP 3: Correlation Analysis
num_cols = ['Annual_Income', 'Loan_Amount', 'Credit_Score',
            'Interest_Rate', 'Age', 'Loan_Tenure_Years']
corr = df[num_cols].corr().round(4)
print(corr.to_string())
print(f"Income <-> Loan Amt      :
      {df['Annual_Income'].corr(df['Loan_Amount']):.4f}")
print(f"Credit Score <-> Int Rate:
      {df['Credit_Score'].corr(df['Interest_Rate']):.4f}")
print(f"Credit Score <-> Loan Amt:
      {df['Credit_Score'].corr(df['Loan_Amount']):.4f}")

# STEP 4: Feature Engineering
def credit_category(score):
    if score < 600:   return 'Low (300-600)'
    elif score < 700: return 'Medium (600-700)'
    elif score < 800: return 'Good (700-800)'
    else:             return 'Excellent (800+)'

df['Credit_Category'] = df['Credit_Score'].apply(credit_category)
print(df['Credit_Category'].value_counts())

# STEP 5: Graph 1 - Distribution of Annual Income
fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(df['Annual_Income'] / 1e6, bins=30,
        color='#1976D2', edgecolor='white', alpha=0.85)
ax.axvline(df['Annual_Income'].mean() / 1e6,
           color='red', linestyle='--', linewidth=2,
           label=f"Mean = Rs {df['Annual_Income'].mean()/1e6:.2f}M")
ax.axvline(df['Annual_Income'].median() / 1e6,
           color='orange', linestyle='--', linewidth=2,
```

```
           label=f"Median = Rs {df['Annual_Income'].median()/1e6:.2f}M")
ax.set_title('Distribution of Annual Income', fontweight='bold')
ax.set_xlabel('Annual Income (Millions Rs)')
ax.set_ylabel('Number of Customers')
ax.legend()
plt.tight_layout()
plt.savefig('graph1_income_distribution.png', dpi=150,
            bbox_inches='tight')
plt.show()

# STEP 6: Graph 2 - Correlation Heatmap
fig, ax = plt.subplots(figsize=(8, 6))
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, annot=True, fmt='.3f', cmap='coolwarm',
            mask=mask, linewidths=0.5, ax=ax,
            annot_kws={'size': 11})
ax.set_title('Correlation Heatmap - Numerical Features',
             fontweight='bold')
plt.tight_layout()
plt.savefig('graph2_correlation_heatmap.png', dpi=150,
            bbox_inches='tight')
plt.show()

files.download('graph1_income_distribution.png')
files.download('graph2_correlation_heatmap.png')
```

Output

Dataset Overview

- **Total Records:** 1000 rows × 12 columns
- **Columns:** Customer_ID, Age, City, Annual_Income, Credit_Score, Loan_Type, Loan_Amount, Interest_Rate, Loan_Tenure_Years, Transaction_Amount, Transaction_Type, Transaction_Date

Descriptive Statistics

Table 1: Key Descriptive Statistics – Numerical Variables

Statistic	Annual Income (Rs)	Loan Amount (Rs)	Credit Score	Interest Rate (%)
Count	1,000	1,000	1,000	1,000
Mean	11,40,413.29	25,66,044.47	595.44	11.03
Median	11,33,202.00	25,84,657.50	596.00	11.05
Std	4,74,802.04	13,94,984.17	171.46	5.48
Min	3,07,531.00	50,175.00	300	1.02
Max	19,99,344.00	49,94,705.00	900	20.99

Correlation Matrix

Table 2: Pearson Correlation Matrix – Numerical Features

	Ann. Income	Loan Amt	Credit Sc.	Int. Rate	Age	Tenure
Annual Income	1.0000	0.0117	0.0226	0.0346	0.0431	0.0475
Loan Amount	0.0117	1.0000	0.0249	-0.0115	0.0251	0.0414
Credit Score	0.0226	0.0249	1.0000	-0.0022	0.0177	-0.0062
Interest Rate	0.0346	-0.0115	-0.0022	1.0000	-0.0136	-0.0096
Age	0.0431	0.0251	0.0177	-0.0136	1.0000	-0.0736
Loan Tenure (Yrs)	0.0475	0.0414	-0.0062	-0.0096	-0.0736	1.0000

Key Highlighted Correlations

- **Annual Income ↔ Loan Amount:** $r = 0.0117$ (near-zero positive)
- **Credit Score ↔ Interest Rate:** $r = -0.0022$ (near-zero negative)
- **Credit Score ↔ Loan Amount:** $r = 0.0249$ (near-zero positive)

Credit Category Distribution

Table 3: Customer Count by Credit Score Category

Credit Category	Count	Score Range
Low	510	300 – 599
Good	173	700 – 799
Excellent	160	800 – 900
Medium	157	600 – 699

Graph 1: Distribution of Annual Income

The histogram below shows the frequency distribution of annual income across all 1000 customers, with mean and median reference lines overlaid to highlight central tendency.

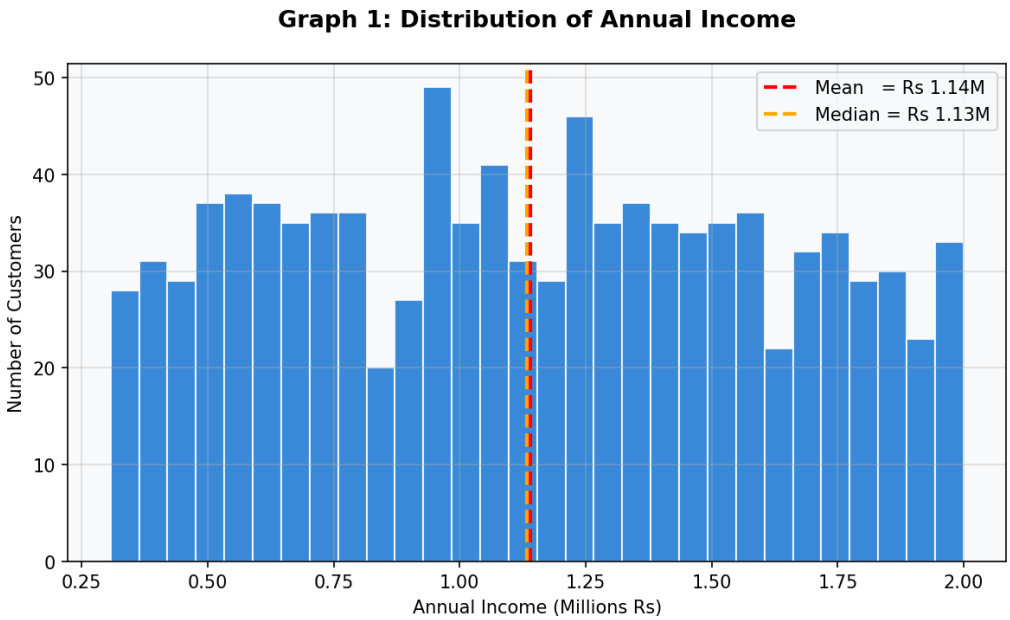

Figure 1: Histogram of Annual Income with Mean and Median Reference Lines

Graph 2: Correlation Heatmap

The heatmap below visualises the Pearson correlation coefficients between all numerical features, supporting data-driven analysis of variable relationships in the BFSI dataset.

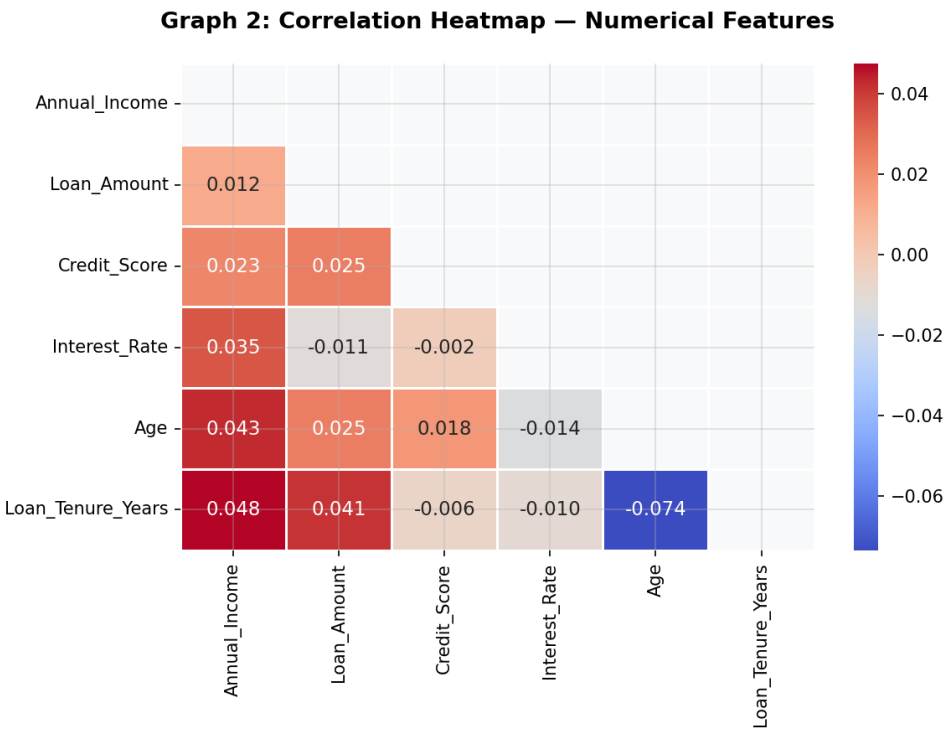

Figure 2: Correlation Heatmap of Numerical Features

Observations

1. **Income Distribution:** The mean annual income (Rs 11,40,413) and median (Rs 11,33,202) are very close, indicating a roughly symmetric income distribution with minimal skew. The standard deviation of Rs 4,74,802 shows moderate variability across customers.

2. **Loan Amount Spread:** The mean loan amount is Rs 25,66,044 with a high standard deviation of Rs 13,94,984 — more than half the mean — indicating a wide range of loan sizes from small personal loans (as low as Rs 50,175) to large home loans (up to Rs 49,94,705).

3. **Credit Score Range:** Credit scores span the full range of 300 to 900 with a mean of 595.44. Over 50% of customers fall in the Low (300–599) category, indicating moderate overall portfolio risk.

4. **Weak Correlations:** All Pearson correlation values between financial variables are close to zero. The near-zero value of $r = -0.0022$ between Credit Score and Interest Rate suggests that this synthetic dataset does not strongly encode real-world risk-based pricing, where lower credit scores are typically penalised with higher interest rates.

5. **Income Histogram:** The histogram confirms a roughly bell-shaped distribution of annual income, with the mean (red dashed) and median (orange dashed) lines nearly overlapping — a hallmark of a symmetric distribution with minimal skew.

6. **Feature Engineering:** Binning credit scores into four categories (Low, Medium, Good, Excellent) enables segment-level analysis and is a standard preprocessing step in BFSI credit models.

Result

Statistical analysis was successfully performed on a banking dataset of 1000 records. Key metrics — mean annual income of Rs 11,40,413, mean loan amount of Rs 25,66,044, and mean credit score of 595.44 — were computed along with a full Pearson correlation matrix. The income distribution histogram confirmed a symmetric distribution, and the correlation heatmap revealed near-zero relationships between all numerical variables in this synthetic dataset. The analysis confirms that descriptive statistics and correlation techniques are fundamental tools for understanding customer financial profiles and supporting data-driven credit decision-making in the BFSI domain.